



API Integration Roadmap

1. Already Connected

Integration	Details	Status
Supabase Auth + Google OAuth	Primary login for users, collectors & admins. Google OAuth via Supabase provider.	✓ Live
Supabase Database	Tables: users, uploads, collectors, drop_points, pickups, notifications	✓ Live
Supabase Storage	Photo uploads to "uploads" bucket. Public URLs generated on upload.	✓ Live
Browser Geolocation	navigator.geolocation captures lat/lng during pickup requests — not yet visualized.	■ Partial
Supabase RPCs	approve_upload, reject_upload, create_daily_settlement	✓ Live

2. Missing APIs — Priority Order

CRITICAL — App incomplete without these

1. Google Maps / Mapbox

CRITICAL

- Admin dashboard has "Interactive map — coming soon" placeholder
- Users submit GPS coords that are never shown on a map
- Drop points have no distance-from-user calculation

Recommendation: Google Maps JS API (map widget) + Places API (address autocomplete for pickup requests)

Effort: 2–3 days of frontend work

2. Razorpay — Payments

CRITICAL

- Collectors are owed money but zero actual payment processing exists
- "Settle" button only updates a DB flag — no real bank transfer happens
- Collector trust breaks down without real payouts

Recommendation: Razorpay Payout API for direct bank transfers to collectors

Effort: 2–3 days (plus KYC / business account setup with Razorpay)

HIGH PRIORITY — Needed for good UX

3. Firebase Cloud Messaging / OneSignal — Push Notifications

HIGH

- Notifications table exists in DB and entries are saved — but never pushed to device
- Users never know when their upload is approved or rejected
- Critical for re-engagement and trust

Recommendation: FCM wired via Supabase Edge Function webhook triggered on approve_upload RPC

Effort: 1–2 days

4. Twilio / MSG91 — SMS

HIGH

- No OTP, no SMS when a collector is assigned to a pickup
- India-critical: collectors and non-smartphone users depend on SMS
- MSG91 has better DLT compliance for India

Recommendation: MSG91 (cheaper for India, supports DLT registration for transactional SMS)

Effort: 1 day

NICE TO HAVE — Makes it smarter

5. Google Vision API / TensorFlow.js — Photo AI

FUTURE

- Fraud detection is currently manual + basic heuristics only
- Could auto-detect if a photo actually contains cans or bottles
- Could auto-count items as a sanity check vs user-reported number

Recommendation: Start with Google Vision API (label detection), move to custom model later

Effort: 3–5 days

6. Resend / SendGrid — Transactional Email

FUTURE

- No welcome email, no weekly GreenCoins summary, no reward redemption receipt
- Retention driver: weekly summaries drive repeat recycling behaviour

Recommendation: Resend (simpler API, generous free tier, great deliverability)

Effort: 1 day

3. Recommended Build Order

#	API	Why first?
1	Maps API	Most visible gap. GPS data already collected, just not shown.
2	Razorpay	Financial backbone — without real payouts collectors disengage.
3	FCM Push	Drives re-engagement after upload approval/rejection.
4	MSG91 SMS	Collectors need SMS; they may not have the app open.
5	Resend Email	Retention tool — weekly summaries boost repeat recycling.

6	Google Vision / AI	Scale milestone — reduces admin review load significantly.
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4. Security Notes — Fix Before Production

- Supabase anon key is hardcoded in src/lib/supabase.js — move to VITE_ environment variables.
- Photo uploads have no file type or size validation on the frontend — add before launch.
- Admin role check happens in UI only — Supabase Row-Level Security (RLS) policies must enforce this at the DB layer too.